



Cape Peninsula
University of Technology

Student: Thabiso Matsaba (220296006)

Course: Project Management 3

Course Code: PRM370S

Lecturer: Mr A. Afolayan

Assignment: Term 3 Individual Research Project

Due Date: 26 August 2022

Total Marks: 150

Title: Agile Project Management in Financial Technology.

Subtitle: Agile Methods and Practises used in the Management of Projects in Financial Technology.

Question: What are the critical factors that influence the success of projects in financial technology implementing agile versus traditional methodologies?

Cape Peninsula University of Technology
Department of Information Technology
Plagiarism Declaration

Subject: Project Management 3 (PRM370S)

Due Date: 26 August 2022

Lecturer: Mr A. Afolayan

I am submitting the assignment for:

☒ an individual project or

☐ a group project on behalf of all members of the group. It is hereby confirmed that the submission is authorized by all members of the group, and all members of the group are required to sign this declaration.

I/We declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) if applicable, the submitted soft copy of the assignment is identical to the hard copy to be submitted.

I/We also acknowledge that I am/we are aware of CPUT policy and regulations on plagiarism and honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the CPUT website

https://www.cput.ac.za/storage/study_at_cput/STUDENT_Rules_Regulations_2019.pdf

In the case of a group project, we are aware that each student is responsible and liable to disciplinary actions should there be any plagiarized contents/undeclared multiple submission in the group project, irrespective of whether he/she has signed the declaration and whether he/she has contributed directly or indirectly to the problematic contents.

It is also understood that assignments without a properly signed declaration by the student(s) concerned and in the case of a group project, by all members of the group concerned, will not be graded by the lecturer(s).

Thabiso Matsaba

220296006



Name

Student ID

Signature

Table of Contents

Agile Software Development in Financial Technology	1
Introduction.....	1
Literature Review	2
Broad View	2
Agile Project Management and its Antecedents.....	2
Traditional Project Management and its Evolution	2
The Strengths and Weaknesses of Agile Project Methodologies.....	3
The Challenges of Implementing Agile Methodologies.....	4
The Impact of Agile Project Management on IT Projects	6
Narrow View.....	7
What are the critical factors that influence the success of projects in financial technology implementing agile versus traditional methodologies?	7
Findings	9
Conclusion	10
Bibliography	11

Agile Software Development in Financial Technology

Introduction

Project management has the aim to deliver projects on time, respecting the initial requirements and the agreed quality, within the planned budget. The achievement of project management is perceived as planning the project accurately at the beginning and then executing the project according to this plan.

Agile project management is not new, but it has increased in popularity in the past few years. Many people believe that using an agile approach to managing projects is different from a more traditional approach, especially as described in the Project Management Institute's Project Management Body of Knowledge(PMBOK® Guide). The Agile mindset changes the way we think and how we approach work in general (Broza, 2016). Agile gives us tools that enable us to manage complex projects today. Projects in many organisations are becoming ongoing development of products, solutions, or services, with increasing emphasis on delivering business value.

The objective of the research is to investigate agile project management in financial technology, primarily focusing on the practises and methods used in agile project management and the critical factors that influence the success of projects in financial technology implementing agile versus traditional methodologies. The literature review will delve into the following subtopics, and closely examine agile project management and its antecedents, traditional project management and its evolution, the strengths and weaknesses of agile project management methodologies, the challenges of implementing agile methodologies, the impact of agile project management on IT projects and advancements and future projections of agile methodologies.

To investigate this further, I have reviewed existing literature that relates and best describes agile project management in financial technology. Specifically focusing on the question: What are the critical factors that influence the success of projects in financial technology implementing agile versus traditional methodologies? I critically analysed each article and related the common themes and arguments to best answer this question.

I will report on the success factors of agile project management. Other project management methods are not the focus of our research. Furthermore, the priority will be upon agile project management opposed to traditional project management

Literature Review

Broad View

Agile Project Management and its Antecedents

The environment in which projects are implemented has become highly dynamic and predictable. The cultural, social, and organizational environments in which projects are implemented are ever changing (Collyer, 2009). Modern projects are also characterized by numerous categories of stakeholders who have a significant impact on the projects. The focus on project management has also shifted from getting projects completed within scope, schedule, and cost to ensuring that project outcomes satisfy customers and all stakeholders (Cervone, 2010). It has also become paramount for project teams to collaborate closely with other stakeholders to increase chances of realizing project success. Consequently, project managers need approaches that are adaptable and flexible.

Agile project management (APM) was developed to respond to this need. Agility refers to the ability to take initiative-taking actions in an unpredictable and ever-changing environment (Xu, 2009). Also, APM is an incremental and iterative approach of managing projects, where project developers and stakeholders are expected to work together to identify the product that needs to be built and prioritize specifications (Adjei & Rwakatiwana, 2009). The goal of APM is to enhance the speed of project delivery by reducing the effect of complexity and uncertainties. There are several critical success factors in agile project management including flexibility, team competence, managerial support, speed, and increased involvement of project users.

Traditional Project Management and its Evolution

Traditional project management (TPM) relies on the plan or process approach to manage projects. TPM approaches are typified by premeditated and well-organized plans and control methods (Lozo & Jovanovic, 2012). The plan approach emphasizes on the development of a detailed plan upon which all activities of the project are developed and implemented. Additionally, the process approach emphasizes the use of five groups of processes: initiating processes, planning, executing, monitoring, and controlling, and closing processes (Sharma et al., 2012). The processes are implemented in a linear fashion. These TPM approaches are found to be effective in projects that have well defined scope and that are implemented in environments that have little complexities and uncertainties. Recently, traditional project

management approaches have been criticized because evidence show that projects are becoming increasingly complex and their environments are becoming highly unpredictable (Karamitsos et al., 2010). The business environment has become very dynamic and customer expectations have become volatile. Consequently, it has become difficult for project managers to rely on the traditional project management approaches that stress the development of rigid plans and adherence to rigid processes.

The Strengths and Weaknesses of Agile Project Methodologies

This literature review will focus on two agile methodologies Scrum and Extreme Programming (XP). Scrum was first described by Jeff Sutherland, Ken Schwaber, and Mike Beedle in 1996. It focuses on Agile project management techniques rather than the development aspects of projects. Scrum is more focused on managerial skills of both managers and developers. The term 'Scrum' is taken from the game 'Rugby' where the player passes the ball in steps to hit the goal, which is similar as moving a project in iterations to meet the core objective.

Scrum is iterative, incremental and the process to develop any project, product or managing any work needs to be done quickly. It is a light-weight software development process consisting of implementing a small number of customer requirements in two-to-four-week sprint cycles. The continuous integration of using small sprints reduces the risks and helps in gaining client confidence. The daily stand-up meeting is an immensely powerful approach to manage and drive the sprint and hence the project . The teams tend to be small and usually comprising of about seven to ten people. The Scrum methodology is aimed at project management in changing environments, and it fits well into small projects and some work requirements can be prioritized in a well-structured manner.

The limitations of the Scrum methodology are that they collaborate well with small teams and therefore growing the team may require extended coordination. The project is highly dependent on the cohesiveness of the team members, a minor lack in coordination or communication may cause major impact in the sprint. Scrum does not explicitly address the issue of criticality. The customer is offsite and tight customer collaboration is not possible. Improved team dynamics enabled by Scrum are not available in one-developer projects.

Extreme Programming (XP) is a well-known and a lightweight discipline of software development that focuses on engineering practices. It was introduced in 1998 by Kent Beck, Ron Jeffries, and Ward Cunningham based on experience from the C3 project.

XP is a system of practices which is intended to improve software quality and quickly addresses the changing customer requirements to meet business needs. XP consists of gathering informal requirements from on-site customers, organizing teams of pair programmers, developing simple designs, continuous refactoring, continuous integration, and testing; and advocates frequent releases in short development cycles that improves productivity as well introduces checkpoints where new customer requirements can be embraced. XP is best suited for projects that require collocated teams of small to medium size team. On the project side XP is meant for projects where the requirements are unstable and unpredictable.

The limitations of XP are that it is not suitable for large, difficult, and complex projects. It requires great amount of coordination between the programmers while doing pair programming and any small conflict may damage the objective of collective code ownership and hence impact the iterations. Development of 'metaphor' is required to be shared within the team carefully to ensure the mutual understanding of the terminology. Pair programming is an especially important practice in XP. However, it cannot be applied to one-developer projects. Customer collaboration is not extraordinarily strong, and testing of the code development is done by the same person. All the foreseeable problems may not be found because the developer tests from the same perception the product is built.

The Challenges of Implementing Agile Methodologies

It may be a challenge to implement an agile approach when factors such as scope and schedule are taken into consideration. Agile methods emphasize the importance of a collaborative, people-oriented approach to software development (Fowler and Highsmith, 2001). Consequently, agile methods ushered in several changes in traditional software development roles and practices.

A significant challenge of agile project management is the difficulty in scheduling projects tasks. The agile approach discourages project teams from developing detailed specification and plans at the initial stages of the project (Kurup & Sidhardhan, 2015). Specifications and plans are made for each cycle during implementation and not for the whole project. This approach makes it difficult for the project team to estimate the duration of the entire project leading to scheduling problems. This problem may have a significant impact on the project given that a project schedule contributes significantly towards planning for the acquisition of project resources (Stare, 2013).

The absence of accurate schedules means that the project team cannot plan for the acquisition of inputs well in advance exposing the project to risks such as price fluctuation. The lack of clarity regarding the completion time of the project also makes it difficult for organizations to plan for other projects. It can be concluded that the major challenge for a company in the use of agile project management is the lack of proper project schedule due to the presence of different small projects in a one project, which may cause increases in overall project cost.

The agile project management approach can also make it difficult for the project team to monitor and control the scope of the project. This methodology can encourage excessive changes to the project requirement resulting in significant increase in cost and time (Leybourne, 2009). At times, it may be difficult for the project team to differentiate between changes that are necessary and nice-to-have changes leading to scope creep. Similarly, the agile approach makes quality assurance and control difficult because of lack of predetermined specifications and plans against which performance can be measured and verified (Kurup & Sidhardhan, 2015).

The project is implemented without clear predefined standards and requirements; hence, the project manager lacks clear guide against which to gauge the performance of the team. It is concluded that lack of complete project scope in the hand of project team do not allow them to evaluate the proper performance of team members to judge that either product development is going in a right direction or not. Agile project management also creates a challenge when it comes to knowledge management. Traditional project management approaches emphasize the use of formal communication channels and a lot of documentation (Sharma et al., 2012).

Subsequently, Agile project management encourages the project team to cut back on structured forms of communication and decision-making. Consequently, a lot of the knowledge in agile project management is tacit and undocumented meaning that it resides in the minds of project team members. Many organizations view such a situation as unacceptable because it shifts the balance of power from the management to the project implementing team (Kurup & Sidhardhan, 2015). The tacit nature of Agile project management knowledge also makes it difficult for the organization to promote transfer of this knowledge across the organization.

The Impact of Agile Project Management on IT Projects

In the last twenty years, development of new products and services is tightly connected with the use of IT technology. The rapid expansion and growth in the use of computers in everyday life and business processes has set up the foundation for a new industry around software development. This is followed with a substantial number of new software development companies competing between each other, creating one competitive environment in which surviving is not easy, especially having in mind the increasing requirements of market.

Agile project management has a significant impact on IT projects, regarding attitude of the developers. The methodology results in a high positive affect level among the developers because it can be introduced and incorporated into the existing knowledge of the team members (Mazni, Sharifah-Lailee & Azman, 2011). The review of literature shows that agile methodology increases collaboration and ownership during software development process. The customer, scrum master, and development teamwork collaborate daily (Mark, 2014). The scrum meetings conducted daily enable the developers to organize the completed and future work to address the challenges. The team demonstrates and discusses the product directly with and other stakeholders. The project reflection meetings allow the developers and customers to regularly reflect on the successes and failures of the project.

The use of agile methodology enables the teams to develop software of high quality (Maruping & Viswanath 2009). The teams gather customer requirements at the initial stage of software development. They struggle to understand the customer needs well to enable them develop software that complies with those requirements. However, the customers' needs usually change over the period of software development because they may not understand their needs clearly at the start. They are also likely to identify additional requirements or change their views about the initially stated needs, thus requiring the developers to make appropriate changes that comply with their needs.

Agile software developers are usually aware that the customer requirements are likely to change during the development process, which enable them to have appropriate measures in place to manage the changes effectively when they occur. The frequent changes in requirements increase flexibility of the teams, which enable them to respond effectively. Flexibility to different changes enables them to develop high quality software in accordance with customer requirements (Fergis, 2012).

Narrow View

What are the critical factors that influence the success of projects in financial technology implementing agile versus traditional methodologies?

The Agile method is very innovative compared to other models. This leads a company to innovation and to consistent work on the product results in developing the latest and effective solution that meets the customer needs and wants. For example, in the case of the increase in the use of smartphones, nowadays, people barely step into the branch for the transaction; they just use their phones. The 2018 U.S. Fintech market report described that nearly half of the respondents used mobile devices at a retail store. Almost 48 percent of people use apps to pay their bills. Financial institutes using the Agile approach can benefit from changing the trends quicker and offer faster services, which will increase the customers. The Agile method proved to be very useful in challenging situations as the Agile development team can work in the cycle to find an innovative and productive solution, which can meet the new demands of the customer, quicker than other traditional software.

Authors Boehm and Turner state that there are crucial factors that lead up to successful implementation of agile methods but denote that there are risks involved, only mitigated when tools and techniques involved in the environment are aligned to adopt this method. Factors that assist in successful implantation of agile approaches within an organization include, availability to dedicated business expertise, tools, and techniques as available to members, agile teams should be measured by overall value and added creation lastly, smaller teams usually succeed in agile settings and members should be allocated to a singular project. Success factors of a project are measured against dimensions which represent the environment and resources used in a project such as, Organizational structure, Human resource, processes, technical and project artefact. (Boehm, B. and Turner, R., 2005, pp.30-39.)

According to Harry Henderson, traditionally, projects were developed by first creating a complete plan agreed on by both the developers and the clients and then building the software according to that plan. There are many problems with the traditional way software is developed, information that was not thought of during the initial plan creation could have been left out and specific details may not have been explained well enough for the

developers to fully understand what they need to create. To correct these overlooked or omitted items, the developer team needs to have another meeting with the clients which, in a lot of cases, costs time, of which neither the client nor the developer have set aside for this and can also cost a lot of money. Customers needed working software quicker and sometimes discovered new requirements that were not anticipated. Agile strategies are a response to these new needs. The agile approach delivers working software a lot sooner than the traditional method as well as going through continuous testing and gathering feedback, so the client gets everything they need. (Henderson, H., 2017)

Elena Zavyalova, Dmitri Sokolov and Antonina Lisovskaya say that traditional project management uses diverse human resource management architectures. While some companies would benefit from using agile approaches, a company should evaluate their project management approach and compare it to their human resource management architecture, this will determine whether an agile or traditional approach suits the company best.

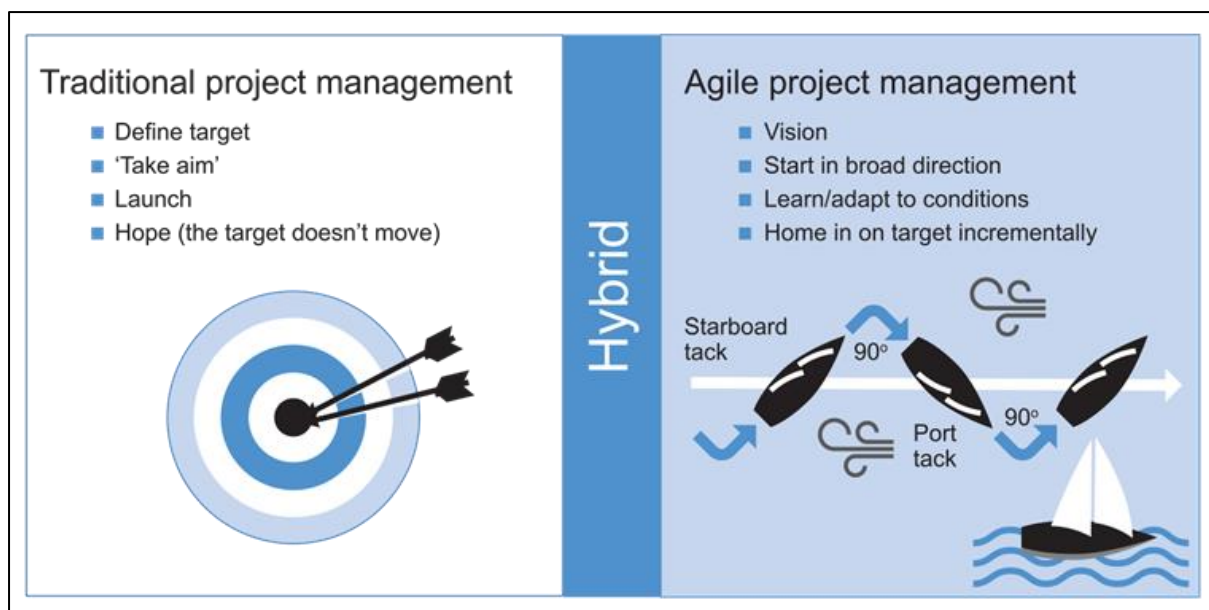
In summary, some experts think the critical flaws of traditional project management are the key attributes that distinguish it from agile. Most researchers above mentioned that where its strengths are is that traditional project management is structured, well-organized, and emphasizes on pre-planning. Traditional project management approaches a project holistically, by planning tasks well ahead and carrying them out sequentially. Traditional project management's structured nature makes it more dependable and methodical. However, research shows that the nature of traditional project management and that of projects, are direct opposites. Projects are dynamic and prone to changes, whereas traditional project management is static, and resistant to changing conditions.

The key factors of traditional project management make it incapable of resolving emerging problems. Instead, the adopting of traditional project management increases pitfalls in a project while agile project management can capitalize on such pitfalls due to its flexibility within projects. Although agile project management can be amazingly effective in many instances, this is dependent on the project manager understanding the agile framework. The rewards of agile management cannot be obtained if the project manager does not have this understanding.

Findings

Agile project management is an iterative way of conveying a project for the duration of its life cycle. There are multiple gradual steps in an agile life cycle towards completing a task. Iterative methodologies are as often as possible utilized in software development to advance speed and versatility since the advantage of iteration is that you can change as you come as opposed to following a direct way. One of the points of a coordinated or iterative methodology is to deliver benefits all through the cycle instead of just toward the end. At the center, agile projects should display focal qualities and practices of trust, adaptability, strengthening, and coordinated effort.

Traditional project management on the other hand, is a widespread practice that incorporates a set of created strategies utilized for arranging, assessing, and controlling exercises. The point of those strategies is to arrive at the ideal outcome on schedule, inside spending plan, and as per requirements. Traditional project management is mostly utilized on projects where activities are finished in a fixed order and there are infrequently changes to be made.



Source: APM-Directing Agile Change

Agile is characterized by both a broad range of techniques and methodologies and a wide array of principles and values that it shares. There are distinct advantages and disadvantages to each of them. The most popular agile frameworks are Scrum and Kanban. Product development projects that are innovative and complex usually require the use of Scrum. However, nothing is perfect, and drawbacks are inevitable. Adopting the Scrum

framework into large projects can be challenging. Only experienced team members can ensure the success of the framework and if one of them leave whilst the project is on the way, it can have a huge negative effect on the project. This does not mean we should frown upon the Scrum framework. Project deliverables can efficiently with Scrum. Daily scrum meetings also allow each team member to express their individual efforts.

Kanban is a highly efficient method of managing software development processes. Compared to Scrum, Kanban is a methodology that is less structured. Processes are not outlined in this methodology. As a model, it introduces a method to improve a process incrementally. There is no specific role prescribed by Kanban. Whenever there is capacity, new items can be added to the board. However, as mentioned before, drawbacks are inevitable. There is no timing associated with each phase, so the process is prone to delays. The development process can be affected by an outdated Kanban board.

Conclusion

Agile project management is a more effective way to manage a project wherein the outcome is not static, and the requirements can change over the project lifetime. This way of managing value client interactions, project adaptability and a functioning result over the traditional values placed on extensive planning and the documentation of processes. However, it is easy to lose track of priorities within a complex enterprise environment where the development of various parts of a system is dispersed over multiple teams and/or concurrent projects.

There is a gap in the knowledge relating to the research question. There are plenty of articles analysing the success factors within traditionally managed projects, however the same cannot be said for projects managed through the agile framework. This information can be acquired from other research about agile, but the number of articles that are solely based on these factors are extremely minimal. Considering this, this literature review still managed to find enough information about the success factors of agile projects from existing research. This area of agile projects should be further researched as it could benefit those seeking to adopt the agile methodology, providing them with core areas that should be focussed on for a greater chance at project success

Bibliography

Agile Project Management on Government Finance Projects

The Journal of Government Financial Management;

Alexandria Vol. 67, Iss. 1, (Spring 2018): 12-18

<https://www.proquest.com/openview/c52e4936305c71ff48196fc4d5c57b39/1?pq-origsite=gscholar&cbl=26015>

Online [08 August 2022]

<https://www.proquest.com/openview/c52e4936305c71ff48196fc4d5c57b39/1?pq-origsite=gscholar&cbl=26015>

Apm.org.uk. 2021. *Agile project management*. [online] Available at:

<https://www.apm.org.uk/resources/find-a-resource/agile-project-management>

[Accessed 24 May 2021].

AGILE SOFTWARE DEVELOPMENT PRODUCTS FOR FINTECH - FINANCIAL TECHNOLOGIES

Dario Gechevski¹ , Klimentina Poposka¹ , Biljana Angelova¹ , Valentina Gecevska²

¹ Ss. Cyril and Methodius University in Skopje, Institute of Economics, Skopje, Republic of Macedonia ² Ss. Cyril and Methodius University in Skopje, Faculty of Mechanical Engineering, Skopje, Republic of Macedonia

Online [Accessed 09 August 2022]

https://www.researchgate.net/profile/Natasa-Krstic/publication/328021879_THE_BUSINESS_IMPLICATIONS_OF_CHATBOTS/links/5c078f68a6fdcc315f9deeeb/THE-BUSINESS-IMPLICATIONS-OF-CHATBOTS.pdf#page=115

A Systematic Study on Agile Software Development Methodologies and Practices

Harleen K. Flora et al, / (IJCSIT) International Journal of Computer Science and Information Technologies, Vol. 5 (3) , 2014, 3626-3637

Online [Accessed 10 August 2022]

<https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.434.2453&rep=rep1&type=pdf>

Crossing the Boundaries – Agile Methods in Large-Scale, Plan-Driven Organizations: A Case Study from the Financial Services Industry

Online [Accessed 15 August 2022]

https://link.springer.com/chapter/10.1007/978-3-319-59536-8_24#citeas

Contenteratechspace.com. 2021.

[online] Available at:

<https://contenteratechspace.com/blogs/top-ten-challenges-of-agile-methodology/>

Online [Accessed 24 May 2021].

Ekanayake, H. and Idar, R., 2021. Traditional Project Management Approach, Industry Challenges and Key Attributes: A Case Study of Construction Project Management in Sri Lanka. *Asia Pacific Journal of Advanced Business and Social Studies*, 5(1), pp.52-64.

AltexSoft. 2021. *Agile Project Management: Best Practices and Methodologies*.

[online] Available at:

<https://www.altexsoft.com/whitepapers/agile-project-management-best-practices-and-methodologies/>

Online [Accessed 24 May 2021].

HRISTOVSKI, T., 2021. *Agile methodologies: Kanban Vs Scrum – Advantages and Disadvantages* – iWConnect. [online] Iwconnect.com. Available at:

<<https://iwconnect.com/agile-methodologies-scrum-vs-kanban-advantages-disadvantages>>

Online [Accessed 24 May 2021].

Impact of Agile Methodology on Software Development

Online [11 August 2022]

<https://www.ccsenet.org/journal/index.php/cis/article/view/44383>

Mrsic, M., 2021. *Traditional Project Management · Activecollab Blog*. [online] ActiveCollab. Available at:

<https://activecollab.com/blog/project-management/traditional-project-management>

Online [Accessed 24 May 2021].

Multi-level agile project management challenges: A self-organizing team perspective

[Journal of Systems and Software](#)

[Volume 117](#), July 2016, Pages 245-257

Online [Accessed 12 August 2022]

<https://www.sciencedirect.com/science/article/abs/pii/S0164121216000807>

Project: [The Benefits and Key Challenges of Agile Project Management under Recent Research Opportunities](#)

Online [Accessed 13 August 2022]

https://www.researchgate.net/publication/316239082_The_Benefits_and_Key_Challenges_of_Agile_Project_Management_under_Recent_Research_Opportunities

The Benefits and Key Challenges of Agile Project Management under Recent Research Opportunities February 2017

The Case for Agile Methodologies against Traditional Ones in Financial Software Projects

European Journal of Business and Management Research www.ejbmr.org

European Journal of Business and Management Research www.ejbmr.org

Online [Accessed 14 August 2022]

<https://ejbmr.org/index.php/ejbmr/article/view/741/411>

[Zavyalova, E., Sokolov, D. and Lisovskaya, A. \(2020\), "Agile vs traditional project management approaches: Comparing human resource management architectures", *International Journal of Organizational Analysis*, Vol. 28 No. 5, pp. 1095-1112.](#)

<https://doi.org/10.1108/IJOA-08-2019-1857>

Online [Accessed 15 August 2022]

Simplilearn.com. 2021. *Advantages and Disadvantages of Scrum*.

[online] Available at:

<https://www.simplilearn.com/scrum-project-management-article>

Online [Accessed 24 May 2021].